Schema-Diverse Control Examples

Experiment index

Question and Context

Was the model recognizing authoring templates without learning the tools' real argument schemas?

Change

Diversify synthetic productive examples and schema usage while keeping the broad 100 ACT / 100 CLARIFY / 100 REFUSE BAYMAX contribution.

The goal was transfer across wording and arguments, not rewriting evaluation cases or accepting malformed outputs.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Schema-diverse	150	42/150 (0.280)	0.560	0.752	0.075	0.190
Schema-diverse	30	5/30 (0.167)	0.533	0.733	0.300	0.200
Schema-diverse / strict replay	150	40/150 (0.267)	0.480	0.672	0.025	0.172
Schema-diverse / strict replay	30	5/30 (0.167)	0.500	0.700	0.300	0.150

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Schema-diverse / strict replay (150 cases)	0.395	0.133
Schema-diverse / strict replay (30 cases)	0.323	0.100

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Schema-diverse: 2026-09-27 UTC, 150 cases.

- Schema-diverse: 2026-09-27 UTC, 30 cases.
- Schema-diverse / strict replay: 2026-09-28 UTC, 150 cases.
- Schema-diverse / strict replay: 2026-09-28 UTC, 30 cases.

Interpretation

Historical full-suite success was 42/150; strict replay measured 40/150. Shift success was 5/30.

Higher boundary wording scores than some ACT-heavy variants did not make this checkpoint a universally better model. Correct routing, valid output and correct arguments remain separate capabilities.

Limits

A template-shift argument mismatch can be a genuine schema error, not a formatting false negative. Scorer changes require independent justification.

The old split/initialization caveats remain. Today's generator implementation cannot by itself recover the exact historical selection.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.